Search in Large Spaces without a Utility Function

What question are we trying to answer?

- Can searching large spaces without a utility function produce results that match or exceed those of human designers or utility-driven search?

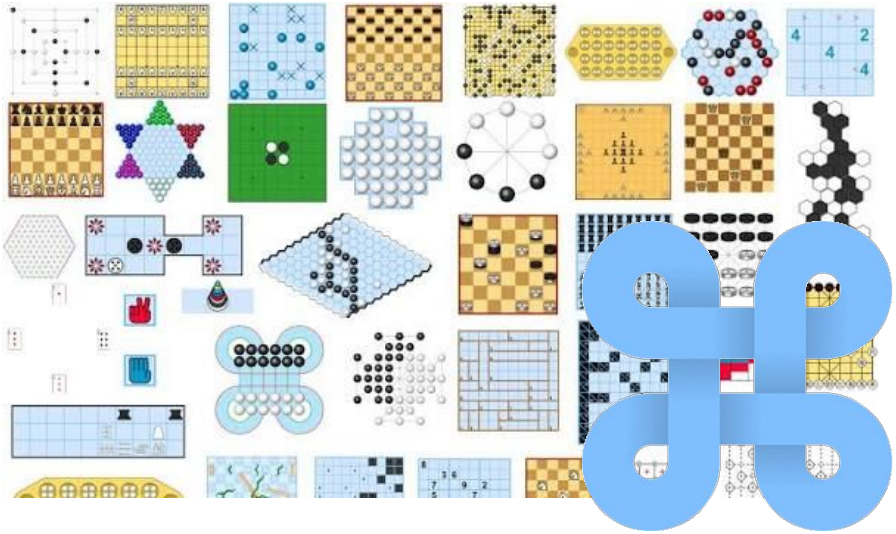
Potential approach: Generating 2D Platformer maps

- Since there are many different variations of 2D platformers having many styles rather than having a utility function may create more creative and unique maps.
- By giving styles, map examples, previous generations we can refine the LLM to create better maps.
- We would have parameters and evaluator to toss the invalid or maps that are not usable and keep the refined ones.
- Ideally have a human factor to critique the maps and help steer the LLM towards the likes and dislikes of what we are looking for.
- In the end compare map generated to human made maps.



Potential approach: Generating games with a GDL

- A GDL is a Game Description Language which can describe a game like chess, checkers, Connect 4, etc.
- The “large space” is all games that can be made with a GDL
- We search the space by having an LLM generate a unique game using the GDL
- There is no utility function that defines what is a fun game
- Several interesting methods could be used to evaluate the game



- Possible issue: How do we get an LLM to actually generate something unique? (not just a minor variation on an existing game)

Something about how would we test it.

Music for Games

Exploring computational creativity without a fixed utility function

Augmented Design Lab · Discussion proposal

2 · Proposed project

Prototype: generate short, looping game-music cues in symbolic form (MIDI).

Control style descriptors: tempo, rhythmic density, repetition, melodic activity.

Compare random/style-constrained, novelty-guided, and human-selected exploration.

Try cues against one simple gameplay scene; no full game required.

3 · Measuring progress

Technical: playable loops, reliable generation, controllable descriptors.

Exploration: diversity and coverage of defined musical feature space.

Human study: listener judgments of fit, distraction, interest, and preference.

Gameplay study (if feasible): task performance or attention with/without music.

Do not assume a single measure of musical “quality.”

1 · Research questions

How can a system explore musical possibilities without a fixed “goodness” score?

How can style be represented: rhythm, texture, harmony, density, instrumentation?

What makes accompaniment interesting without distracting from play?

Can feedback and changing gameplay context guide creative exploration?

4 · Reading

Lehman & Stanley (2011): Abandoning Objectives — novelty search.

Mouret & Clune (2015): Illuminating Search Spaces by Mapping Elites.

Pearce & Wiggins (2001): Towards a Framework for the Evaluation of Machine Compositions.

Carnovalini & Rodà (2020): Computational Creativity and Music Generation Systems.

Collins (2008): Game Sound — adaptive audio and game context.

First-quarter scope & decision

First milestone: 10–20 short loops plus a simple listening/comparison interface.

Second milestone: novelty/coverage measurements and a small pilot listener study.

Research contribution: what exploration mechanisms help designers discover useful music?

Open question for the team: Which style descriptors and evaluation tasks are most informative?